LabTech Testing Limited

STANDARD TEST PROCEDURE

ASTM C940
Expansion and Bleeding of Freshly Mixed Grouts for Preplaced-Aggregates Concrete

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Table of Content

1 Scope 3
2 Terminology 3
3 Equipment 3
4 Test Sample 4
5 Test procedure 4
6 Calculations 5
7 Record 5
8 Report 5
9 Measurement Uncertainty 6

REVISION LEDGER

Rev. No.	Section No.	Page(s)	Revision Outline
1	NA	NA	First Issue
2	3	3	Equipment requirement updated
	5	4	Test procedures updated
	8	5	Report information updated

1 Scope

1.1 This test method determines the amount of expansion and accumulation of bleed water at the surface of freshly mixed hydraulic-cement grout.

1.2 Based on General Specification for C.E Works (2006).

1.3 Based on ASTM C940-22.

2 Terminology

2.1 Bleeding – The process where water rises to the surface of freshly placed grout, creating a layer of excess water.

2.1.1 Bleed Water: The water that separates from the grout mixture and appears on the surface.

2.1.2 Bleed Rate: The speed at which bleed water is expelled, typically measured in milliliters per hour.

2.1.3 Bleedable Grout: Grout formulations that permit some bleeding, which can affect final strength and surface quality.

2.2 Free Expansion: The increase in volume of grout due to hydration or temperature changes, occurring without constraints.

2.2.1 Thermal Expansion: Expansion resulting from temperature increases, which can impact the grout's structural integrity.

2.2.2 Hydration: The chemical process where water interacts with cementitious materials, leading to hardening and volume changes.

2.2.3 Expansion Joint: A designed gap in structures to accommodate the free expansion of materials, including grout.

3 Equipment

3.1 Glass or Perspex container 100±10mm in diameter and a scale readable to 1mm fixed to the outside.

3.2 The apparatus used should be 1000mL.

3.3 Sample size of 800mL should be used.

3.4 Thermometer readable to 0.5℃.

3.5 Sample container minimum 5L.

3.6 Stopwatch.

4 Test Sample

4.1 Samples shall be obtained and tests commenced within one hour after the grout has been mixed, the sample shall be protected from changes in moisture content before the tests for the amount bleeding and free expansion are performed.

4.2 Store samples in a climate-controlled room with stable temperature and humidity levels.

4.3 Use airtight containers to minimize exposure to ambient moisture. Ensure that containers are clean and dry before placing samples inside.

5 Test procedure

5.1 Start volume measurements within 3 min of completion of mixing.

5.2 Maintain the ambient temperature of the room in which the test is performed at 23 $\pm$2°C.

5.3 Measure the temperature of grout in accordance with ASTM C1064.

5.4 Immediately after the completion of mixing, measure the temperature of the grout.

5.5 Place the beakers on a level surface free of vibration. Cover to prevent evaporation of the bleed water.

5.6 Divide each sample into three specimens and pour the grout of 100 $\pm$5 mm of each specimen in a covered beaker with a diameter of 100 $\pm$10 mm.

5.7 Take and record the volume (H_1), to the nearest 1mm. the grout temperature and the time at which the reading made.

5.8 Cover the beakers with an airtight seal.

5.9 Record the readings, to the nearest 1mm, of the upper surface of the grout (H_3) and the upper surface of any bleed water (H_2) at 15 minutes intervals for the first 60 minutes.

5.10 Continue to take readings at hourly intervals until two successive reading no further expansion or bleeding.

5.11 Record and reading of the upper surface of the grout and the upper surface of any bleed water (H_w).

6 Calculations

6.1 Expansion % = $\frac{H_3 - H1}{H_1} \times 100$

6.2 Bleeding % = $\frac{H_3 - H_2}{H_1} \times 100$ at prescribed intervals

6.3 Free Expansion % = $\frac{H_2 - H_1}{H_1} \times 100$

6.4 Final bleeding % = $\frac{H_w}{H_1} \times 100$

6.5 H_1 = volume of sample at beginning of test, mL,

H_2 = volume of sample at prescribed intervals, measured at upper surface of water layer, mL,

H_3 = volume of grout portion of sample at prescribed intervals, at upper surface of grout, mL,

H_w = volume of decanted bleed water, mL.

7 Record

7.1 Worksheet: FST04-GTT-C940

7.2 Test Report template: GTT-C940-RT

7.3 Daily Record: FQS09

8 Report

8.1 Identification of the grout sample.

8.2 Expansion of grout to the nearest 0.2% for each prescribed interval.

8.3 Bleeding of grout to the nearest 0.2% for each prescribed interval.

8.4 Combined expansion of grout plus bleed water to the nearest 0.2% for each prescribed interval.

8.5 Final bleeding to nearest 0.2%.

9 Measurement Uncertainty

9.1 Base on ISO/IEC Guide 98-3 (GUM)

9.2 Step1-Calculate the Type A Uncertainty, you need to perform five repeated measurements of bleeding volume (in mL) and then calculate the mean volume.

9.3 Step2- Calculate the Standard Deviation:

$$s = \sqrt{\frac{\sum(xi - \bar{x})^{2}}{n - 1}}$$

9.4 Step 3- Type B Uncertainty:

9.4.1 Assume the manufacturer specifies that the uncertainty of the measuring device (e.g., graduated cylinder) is ±2 mL.

9.4.2 Therefore, the Type B Uncertainty uB is 2 mL.

End of Procedure